

HOMEWORK 4

Let R be a Dedekind ring and $P \subset R$ a nonzero prime ideal. Prove that the valuation ring of the associated discrete valuation v_P is equal to the localization R_P of R with respect to the prime ideal P .

1. Prove that a submodule of a finitely generated projective module over a Dedekind ring is projective.
2. Let M be a nonzero torsion free module over a Dedekind ring R and $I \subset R$ an ideal. Show that there is an injective R -module homomorphism $I \rightarrow M$.
3. Let I be a fractional ideal of a Dedekind ring R . Prove that $I \oplus I^{-1} \simeq R^2$. Deduce that I is generated by two elements.
4. Let I be a fractional ideal of a Dedekind ring R . Prove that $I^{-1} \simeq \text{Hom}_R(I, R)$.
5. Let $M_1, M_2, \dots, M_n$ be (left) R -modules and $M = M_1 \oplus M_2 \oplus \dots \oplus M_n$. Prove that the ring $\text{End}_R(M)$ is isomorphic to the ring of $n \times n$ matrices (f_{ij}) with $f_{ij} \in \text{Hom}_R(M_j, M_i)$.
6. Let R be a ring. Show that every (left) R -module is free if and only if R is a division ring.
7. Let R be a ring, $S = M_n(R)$. We view R^n as a left S -module. Prove that every S -module endomorphism f of R^n is of the form $f(x) = xa$ for some fixed $a \in R$. Deduce that $\text{End}_S(R^n) \simeq R^{op}$.
8. Let V be a nonzero finite dimensional left D -module over a division ring D . Prove that the ring $\text{End}_D(V)$ is simple and semisimple.
9. Let V be a finite dimensional left D -module over a division ring D and $W \subset V$ a submodule. Prove that

$$\text{Hom}_D(V, W) \simeq \{f \in \text{End}_D(V) \text{ such that } \text{Im}(f) \subset W\}$$

is a right ideal in $\text{End}_D(V)$. Prove that every right ideal in $\text{End}_D(V)$ is equal to $\text{Hom}_D(V, W)$ for a unique submodule $W \subset V$.

10. Let D be a division ring. Prove that the center of D is a subfield of D .